

Kai-Wen Zhao

Senior AI Engineer

Taiwan • 0932234097 • kevin.zhao.dev@gmail.com

Summary

Senior AI Engineer and Tech Lead with extensive experience integrating high-performance surveillance AI across multiple NPIs for cameras and edge appliances. Experienced in guiding multi-site technical teams through collaborative leadership to ensure the delivery of edge AI features on embedded systems.

Experience

AI Tech Lead / Senior AI Engineer, 12/2023

Ubiquiti Inc., Taipei City, Taiwan

- Orchestrated the AI engineering for 10+ NPIs across the AI Series Cameras (G6/Pro) and edge appliances (AI-Key, AI-Port), guiding a multi-site, distributed team of 8 developers to deliver integrated AI solutions.
- Delivered embedded AI functions for cameras, including Object Detection, Face Recognition, License Plate Recognition (LPR), and Audio Event Detection. Architected advanced AI applications, implementing Natural Language Search, Video Summarization, and Person Re-Identification (ReID) to enhance surveillance intelligence.

Computer Vision Developer, 01/2022 - 06/2024

Ubiquiti Inc.

- Engineered a Unified Inference Framework that decoupled model logic from hardware backends, enabling high-performance deployment across diverse NPUs, including Ambarella, SigmaStar, Qualcomm, MediaTek and Nvidia.
- Developed a Face Recognition system from scratch, covering end-to-end pipelines from detection and landmarking to feature extraction and attribute estimation.
- Implemented model compression techniques, such as Pruning and Quantization-Aware Training (QAT), to adapt SOTA models (e.g., YOLOv6) for resource-constrained IoT devices.

Senior Software Engineer, 10/2019 - 12/2021

CyberLink Corp., Taiwan

- Architected facial recognition models for NIST (FRVT) submissions, focusing on deep neural network architecture design and high-precision optimization.
- Achieved a 20% relative reduction in FNMR at a 1e-6 threshold, significantly enhancing the accuracy of lightweight face recognition models for high-security applications.
- Optimized mobile-based face detectors, delivering a 10% inference speed-up and a 6% mAP improvement through architecture refinement and efficient layer design.
- Guided a technical task force to win 3rd place (out of 250 teams) in the ECCV 2020 GigaVision Challenge for pedestrian and vehicle detection.
- Developed and deployed real-time person detection systems for crowded scenarios, ensuring high-throughput execution on GPU platforms.

Machine Learning Engineer, 09/2018 - 10/2019

Viscovery, Taiwan

- Engineered a Vision-based smart checkout system for retail environments, outperforming existing solutions by 15% in verified accuracy.
- Built extensible internal evaluation frameworks for object detection and metric learning, standardizing the R&D workflow and benchmarking process across multiple projects.
- Streamlined end-to-end ML pipelines by optimizing data pre-processing and post-processing workflows, significantly reducing model development cycles.

Research Assistant, 03/2016 - 08/2018

Market Intelligence & Consulting Institute, MIC, Taiwan

Conduct research and synthesize industry background insights across electric vehicles, quantum computing, and renewable energy.

Research Assistant, 08/2015 - 12/2015

National Taiwan University, Taipei City, Taiwan

Research Assistant, 08/2014 - 08/2015

Academia Sinica, Taiwan, Taipei City, Taiwan

- Developed a novel CNN-based strategy for facial attribute extraction using C++, achieving up to 70% relative improvement over benchmarks. Received the Top 10% Paper Award at IEEE MMSP 2015.
- Deployed demographic prediction models (age/gender) using the Caffe framework on Android platforms for commercial audience analytics.
- Engineered 'HeartBus,' a first-aid training system utilizing Microsoft Kinect and Intel RealSense for real-time human motion sensing and interactive CPR training.

Education

Master of Science (M.S.), Physics, 2016 - 2018

National Taiwan University, Taiwan

Published research in Physical Review E titled '*Generation of ice states through deep reinforcement learning*,' demonstrating the application of DRL to complex statistical mechanics problems.

Bachelor of Science (B.S.), Mechanical Engineering, 2009 - 2014

National Cheng Kung University, Taiwan

Skills

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|---------------------------------|----------------------------------|
| • Deep Learning Computer Vision | • Languages: C, C++, Python |
| • Frameworks: Pytorch, TensorRT | • Tools: Docker, Git, DVC, CI/CD |

Achievements

First prize of Taiwan Physics Society (TPS) poster competition (condensed matter group), Top 10% Paper Award from IEEE MMSP, Second prize of Matlab Competition: Solving Traveling Salesman Problem with GPU and Implement a Real Scenario, The Award of Outstanding Student for the Academic Achievement, Professor Ke-Yang Lee Scholarship